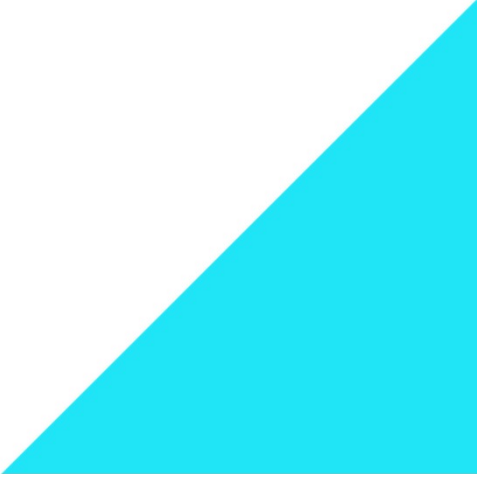
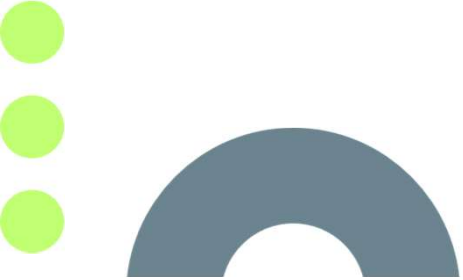


ANOMALY DETECTION IN WASTE MANAGEMENT: A FRAMEWORK

Sara Oliveira, Shazia Tabassum, António
Pinheiro, João Gama, Pedro Santana



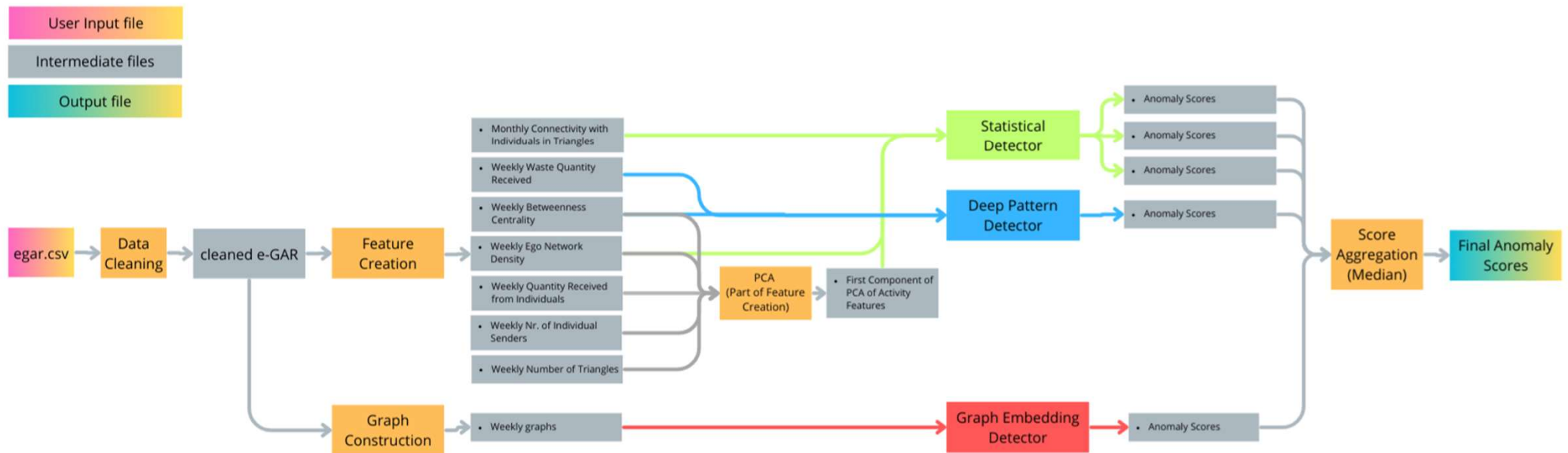
OBJECTIVES

Develop a data-driven tool to identify waste operating companies engaged in suspicious activity, to be flagged for inspection.

- **Develop features** representing relevant aspects of the network and operational behavior of the companies
- Detect **temporal deviations or unusual complex patterns** in the **network and operational activity** of the companies
- **Combine different approaches** to detect different types of anomalous behaviour.
- Obtain an **overall anomaly score** for all waste operating companies.

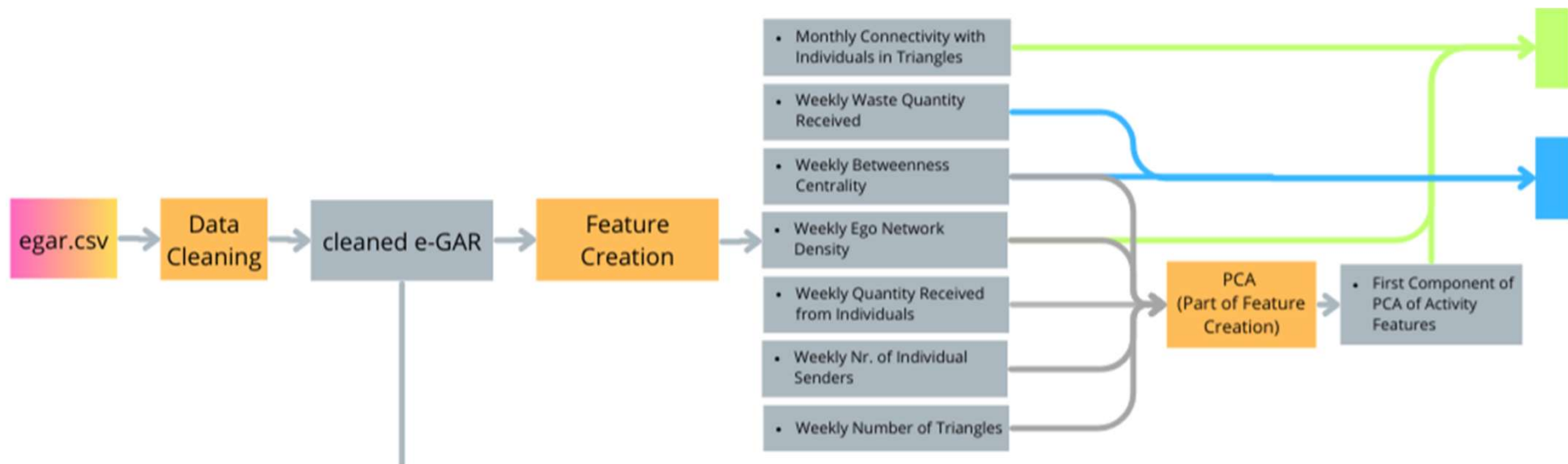


OVERVIEW OF THE FRAMEWORK



FEATURE CREATION:

EXTRACTING INFORMATION FROM THE DATA



STATISTICAL DETECTOR: DETECTING SUDDEN CHANGES IN BEHAVIOR

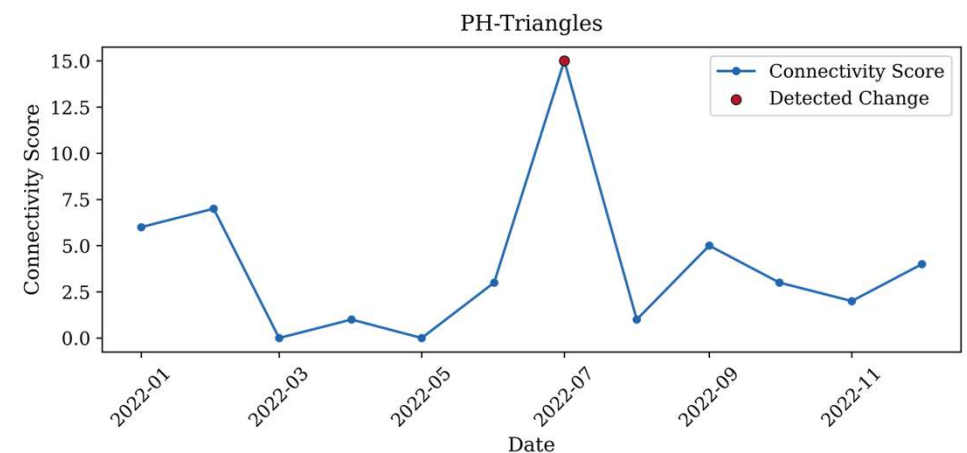
Page-Hinkley test

- Detects **abrupt changes** in the mean of a signal by tracking the **cumulative deviation** from its historical average.
- A **change is detected** when the cumulative sum exceeds the defined threshold.

Applied to the **standardized** features:

- Ego network density (weekly)
- First component of PCA of 5 features (weekly)
- Connectivity with individual senders in triangles (monthly)

The **company score** is defined as the sum of the 3 highest detected change strengths.



DEEP PATTERN DETECTOR: DETECTING UNUSUAL TEMPORAL PATTERNS

Long Short-Term Memory Variational Autoencoder (LSTM-VAE)

- Deep learning model for learning **complex, non-linear patterns over time**.
- The **variational autoencoder** learns a compressed representation of normal activity and the **LSTM component** captures long-term dependencies and temporal dynamics in the companies' behavior.
- The model is trained to minimize **Reconstruction Error + β · Kullback-Leibler Divergence**.

The model was **trained on 3 weekly features**:

- Connectivity with individual senders in triangles
- Betweenness centrality
- Quantity received from individuals

The **company anomaly score** was defined as the mean of 10 highest reconstruction errors.

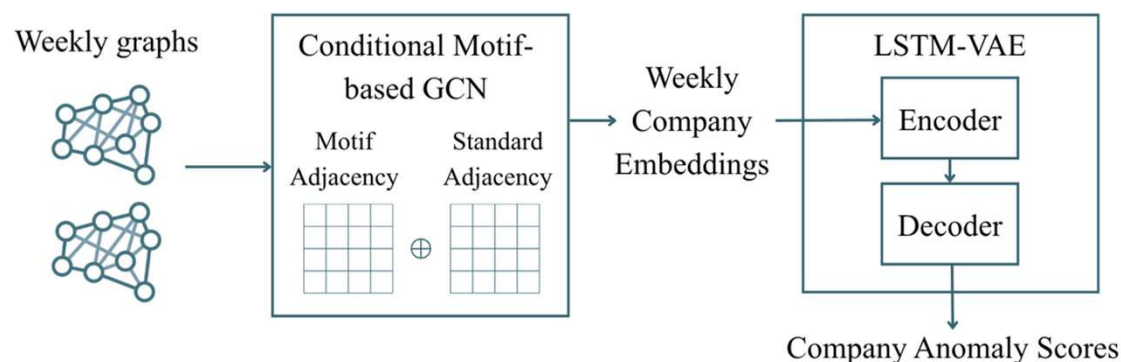
GRAPH EMBEDDING DETECTOR: DETECTING SUSPICIOUS NETWORK STRUCTURES

Conditional Motif-aware Graph Convolutional Network (CM-GCN)

- A **GCN** is a neural network specifically designed for graph data, which learns company representations by **aggregating information from its neighbors**.
- The **CM-GCN** goes beyond **standard pairwise connections**, to learn higher-order connectivity patterns. Particularly, the model captures the **connectivity with triangle motifs including nodes-of-interest** (individual waste senders).

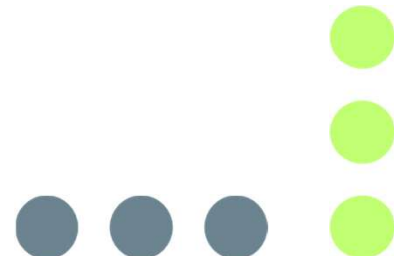
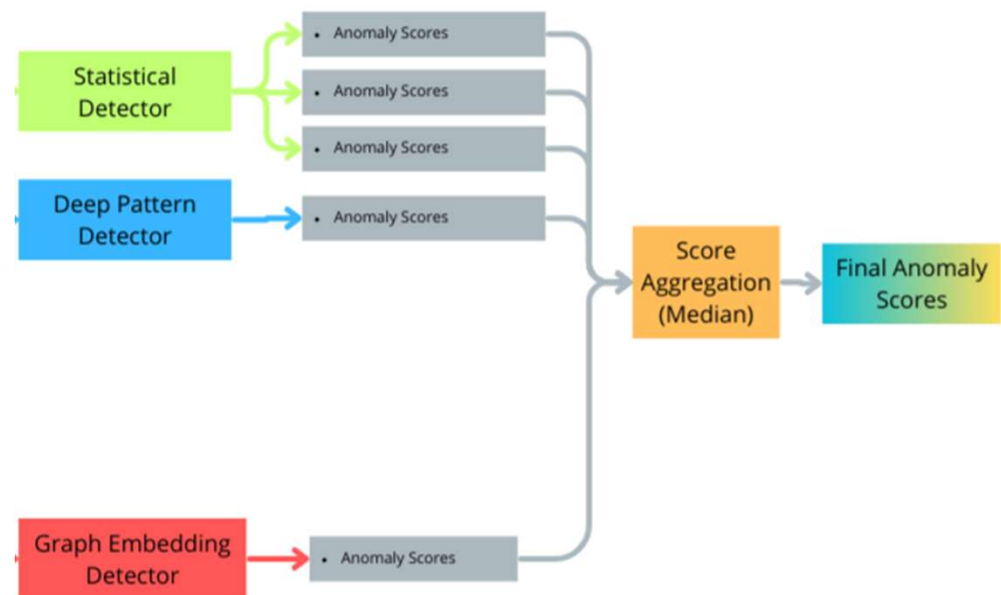
The CM-GCN model generates **weekly company embeddings**, which are passed into an LSTM-VAE model to detect anomalies.

The **anomaly score** was defined as the mean of 10 highest reconstruction errors.



ANOMALY SCORING: COMBINING RESULTS FROM MULTIPLE DETECTORS

- The individual scores from the different methods were **ranked** to ensure comparability, and then **standardized** to obtain scores between 0-1 (1=most anomalous).
- Final anomaly scores: **median of the 5 scores.**
- This approach was chosen due to the **complementarity of the methods.**



REFERENCES

- Oliveira, S., Tabassum, S., Gama, J., Santana, P. (2025). Detecting suspicious activities in waste transport data via temporal and network-aware change detection. In: Machine Learning and Knowledge Discovery in Databases. Workshops Track: European Conference, ECML PKDD 2025, Porto, Portugal, September 15–19. vol. in press. Springer (2025)
- Oliveira, S., Tabassum, S., Gama, J., Santana, P., Garcia, A. (2025). Conditional Motif-based Graph Convolutional Network for Anomaly Detection in the Waste Management Network. Submitted for Symposium on Intelligent Data Analysis (IDA 2026), Leiden, Netherlands, 22-24 April 2026.
- dos Santos Oliveira, S. L. (2025). Detecting Suspicious Activities in Waste Transport Data via Temporal and Network-Aware Change Detection.

